Project Gaja-Saviya

An Intelligent Multi-Modal Framework for Human-Elephant Conflict Mitigation

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Individual Project Proposal Report

Component: Bio-Acoustic Behavior Analyst

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Human-Elephant Conflict (HEC) in Sri Lanka continues to claim lives and livelihoods across rural farming communities. A critical limitation of current mitigation systems is their reliance on presence detection alone, without any understanding of elephant behavioral context. This research proposes the Bio-Acoustic Behavior Analyst, a deep learning-based audio processing component within Project Gaja-Saviya, designed to detect elephant presence through environmental sound analysis and classify behavioral states as aggressive or calm. The component captures environmental audio via a high-sensitivity microphone attached to the IoT Sentinel Node, processes audio frames using the Librosa library, and converts them into Mel-spectrogram representations. A Convolutional Neural Network (CNN) trained on datasets from the Elephant Listening Project and Elephant Voices repositories classifies the behavior. Temporal smoothing is applied across prediction windows to stabilize outputs and reduce noise-induced misclassifications. The component produces structured outputs including a behavior label and confidence score, which are forwarded to the Cross-Modal Sensor Fusion engine for joint threat validation. This behavior-level detection capability, as opposed to simple presence detection, adds a critical layer of situational awareness that reduces unnecessary panic and enables more targeted responses to genuine threats.

Keywords: Bio-Acoustics, Elephant Behavior Classification, Mel-Spectrogram, CNN, Deep Learning, HEC, IoT

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
IoT	Internet of Things
HEC	Human-Elephant Conflict
CNN	Convolutional Neural Network
ML	Machine Learning
MFCC	Mel-Frequency Cepstral Coefficients
PAM	Passive Acoustic Monitoring
ReLU	Rectified Linear Unit
GPU	Graphics Processing Unit
API	Application Programming Interface
ESP32	Espressif Systems 32-bit Microcontroller
FFT	Fast Fourier Transform
SNR	Signal-to-Noise Ratio

1. INTRODUCTION

1.1 Background and Context

Human-Elephant Conflict is one of Sri Lanka's most pressing conservation and rural development challenges. Elephants regularly enter agricultural zones during the night, breaking through fences and destroying crops that represent entire seasons of farmer investment. The psychological toll of constant fear and recurring economic loss drives farmers to take retaliatory actions against elephants, worsening the conservation outlook for an already endangered population.

Current detection systems provide binary alerts — an alarm either triggers or it does not — without offering any insight into whether the detected animal is calmly browsing near the perimeter or actively charging through. This distinction is critically important. A calm elephant near a fence may warrant monitoring and a gentle deterrent, while an aggressive elephant actively breaching the perimeter demands an immediate, strong response and emergency navigation guidance for farmers. Without behavioral context, farmers receive identical alerts for radically different situations, leading to poor response calibration and further erosion of trust in the system.

The Bio-Acoustic Behavior Analyst component of Project Gaja-Saviya addresses this gap directly. By leveraging passive acoustic monitoring (PAM) techniques and deep learning, the component analyses environmental soundscapes to detect elephant vocalizations and classify the behavioral state of the animal. This contextual information is then passed to the Cross-Modal Sensor Fusion engine, where it is combined with electrical fence signal data to produce a holistic, validated threat assessment.

1.2 Literature Review

Passive acoustic monitoring has a well-established track record in wildlife research. Studies by Payne et al. demonstrated that elephant vocalizations carry rich behavioral information — rumbles, roars, trumpets, and infrasound signals each correspond to distinct behavioral states and social contexts. The Elephant Listening Project at Cornell University has produced extensive audio datasets capturing these calls in the wild, enabling machine learning approaches to automated classification.

Deep learning, particularly Convolutional Neural Networks, has transformed audio classification. The breakthrough insight by LeCun et al. — that CNNs can learn hierarchical spatial features — was extended to audio by treating spectrograms as 2D images. Work by Salamon and Bello (2017) demonstrated that CNNs applied to Mel-spectrograms outperformed traditional machine learning methods (SVM, random forests) on environmental sound classification tasks by significant margins.

Mel-spectrograms have become the standard feature representation for audio classification tasks because they compress audio into a perceptually meaningful time-frequency

representation that aligns with the frequency sensitivity of biological hearing systems. Libraries such as Librosa provide robust, well-tested implementations of Mel-spectrogram computation that are suitable for embedded and server-side deployment.

Transfer learning approaches using models pre-trained on large audio datasets (such as VGGish pre-trained on AudioSet) have shown strong performance when fine-tuned on domain-specific tasks with limited data — a critical consideration given the relative scarcity of labeled elephant vocalization datasets. Temporal smoothing via majority voting or moving average across prediction windows has been shown to significantly improve prediction stability in streaming audio classification scenarios.

1.3 Research Gap

Despite the availability of elephant audio datasets and proven deep learning architectures, several gaps remain. Most existing elephant sound classification research focuses on species identification or presence detection rather than behavioral state classification. No published system has implemented real-time behavioral classification integrated within an IoT-based HEC alert framework. Furthermore, audio datasets specific to Sri Lankan elephants — whose vocalizations may differ acoustically from African elephant datasets that dominate the literature — are extremely limited. The challenge of robust classification in noisy tropical forest environments, with interference from rain, wind, insects, and other wildlife, has not been specifically addressed in the HEC context. This research directly targets all of these gaps.

2. OBJECTIVES

2.1 Main Objective

To develop a real-time bio-acoustic analysis system that detects elephant presence from environmental audio captured by the IoT Sentinel Node and classifies elephant behavioral states (aggressive vs calm) using deep learning, providing structured behavioral outputs to the sensor fusion engine.

2.2 Specific Objectives

1. To collect and curate a labeled audio dataset combining Elephant Listening Project, Elephant Voices, and locally collected Sri Lankan elephant vocalization samples.
2. To implement audio preprocessing and Mel-spectrogram feature extraction using the Librosa library, optimized for real-time streaming audio input.
3. To design and train a CNN model capable of classifying audio frames into elephant behavioral categories (aggressive, calm, absent) with accuracy above 85%.
4. To apply transfer learning using a pre-trained audio classification backbone to improve performance on the limited Sri Lankan elephant audio dataset.
5. To implement temporal smoothing across consecutive prediction windows to reduce noise-induced prediction instability in streaming contexts.
6. To design a structured output schema (behavior label + confidence score + timestamp) compatible with the Cross-Modal Sensor Fusion API.
7. To evaluate model performance using precision, recall, F1-score, and confusion matrix on a held-out test set.

3. METHODOLOGY

This research follows the Design Science Research methodology, producing a functional prototype of the Bio-Acoustic Behavior Analyst through iterative design, construction, and evaluation cycles.

Phase 1: Dataset Collection and Curation

Audio data will be collected from three sources. The Elephant Listening Project (Cornell University) provides recordings of forest elephants with behavioral annotations. The Elephant Voices database offers complementary vocalization recordings with behavioral context. A targeted field data collection campaign will be conducted at wildlife monitoring stations in Sri Lanka to capture local elephant vocalizations. All recordings will be annotated with labels: aggressive, calm, or non-elephant (background noise). Data augmentation techniques including pitch shifting, time stretching, and additive Gaussian noise will be applied to address class imbalance and improve model robustness.

Phase 2: Audio Preprocessing and Feature Extraction

Raw audio captured by the Sentinel Node microphone will be segmented into overlapping frames of 1-second duration with a 0.5-second hop. Each frame will be converted to a Mel-spectrogram using Librosa with parameters: sample rate 22,050 Hz, FFT window size 2048, hop length 512, and 128 Mel filterbanks. The resulting spectrograms (128 x time bins) will be normalized using per-sample mean subtraction and standard deviation scaling before being passed to the CNN.

Phase 3: Model Development and Training

A CNN architecture will be implemented using TensorFlow/Keras. The architecture will consist of three convolutional blocks (Conv2D + BatchNorm + MaxPool + Dropout), followed by a GlobalAveragePooling layer and a Dense classification head with softmax activation for three-class output. Transfer learning will be explored using VGGish pre-trained on Google's AudioSet, with fine-tuning on the curated elephant dataset. Models will be trained with Adam optimizer, categorical cross-entropy loss, and early stopping on validation loss.

Phase 4: Temporal Smoothing and API Integration

A sliding window majority vote will be applied across 5 consecutive 1-second predictions to produce a stabilized behavioral label. The smoothed output, paired with the mean confidence score from the window, will be serialized into a JSON prediction object and dispatched to the Cross-Modal Sensor Fusion API via HTTP POST. A Flask/FastAPI server will host the model inference pipeline on the backend server.

4. SYSTEM ARCHITECTURE

4.1 Component Position

The Bio-Acoustic Behavior Analyst operates as an upstream producer feeding into the Cross-Modal Sensor Fusion engine. It receives raw audio data from the Sentinel Node via MQTT, processes it server-side, and dispatches structured behavioral predictions via REST API to the fusion component.

4.2 Internal Sub-Components

- Audio Receiver: Subscribes to MQTT topic for raw audio frames from ESP32 Sentinel Node.
- Preprocessor: Segments, normalizes, and converts raw audio to Mel-spectrograms.
- CNN Inference Engine: Loads trained model and runs per-frame classification.
- Temporal Smoother: Applies sliding window majority vote across consecutive predictions.
- Output Serializer: Formats and dispatches structured prediction JSON to the fusion API.
- Database Logger: Persists raw audio metadata, spectrograms, and prediction history.

4.3 Data Flow

Sentinel Node Microphone -> MQTT Broker -> Audio Receiver -> Preprocessor -> CNN Inference Engine -> Temporal Smoother -> Output Serializer -> Fusion API + Database

5. USER REQUIREMENTS

5.1 Stakeholders

- Primary Users: Farmers receiving downstream alerts influenced by this component's behavioral classification.
- Fusion Component: Cross-Modal Sensor Fusion engine that consumes this component's prediction outputs.
- System Administrators: Technical staff monitoring model performance and triggering retraining.
- Wildlife Conservationists: Secondary beneficiaries of data collected by this component.

5.2 Functional Requirements

8. The system shall continuously receive audio frames from the Sentinel Node microphone via MQTT.
9. The system shall convert received audio into Mel-spectrogram representations for model inference.
10. The system shall classify each audio frame as aggressive elephant, calm elephant, or non-elephant.
11. The system shall apply temporal smoothing across consecutive frames to stabilize predictions.
12. The system shall output a structured JSON object containing behavior label, confidence score, and timestamp.
13. The system shall dispatch prediction outputs to the Fusion API within 3 seconds of receiving audio input.
14. The system shall log all predictions and audio metadata to the backend database.
15. The system shall support model hot-swapping to enable retraining without service interruption.

5.3 Non-Functional Requirements

Requirement	Specification
Accuracy	Model classification accuracy above 85% on test set
Latency	Prediction dispatched within 3 seconds of audio receipt
Noise Robustness	Maintain >80% accuracy at SNR >= 10dB
Availability	99% service uptime for inference server
Scalability	Support 20 concurrent Sentinel Node streams
Maintainability	Modular pipeline enabling model updates without full redeployment

6. BUDGET AND JUSTIFICATION

Category	Amount (LKR)
GPU Cloud Training (AWS/GCP)	22,000
Audio Recording Equipment	12,000
Software Tools & Libraries	5,000
Dataset Storage & Management	4,500
Field Data Collection Travel	6,000
Contingency	3,500
TOTAL	**53,000**

GPU cloud compute is the largest expense due to the training requirements of CNN models on audio spectrogram data. Field recording equipment is essential to collect Sri Lankan-specific elephant audio. Storage costs cover the curated dataset and logged prediction history. Travel expenses support field data collection sessions at wildlife monitoring stations in elephant conflict zones.

7. WORK BREAKDOWN STRUCTURE

Phase	Task	Duration
1	Literature Review & Proposal Preparation	Months 1-2
2	Dataset Collection (Public + Local Field Recording)	Months 2-4
3	Audio Preprocessing & Mel-Spectrogram Pipeline	Months 3-4
4	CNN Model Design & Initial Training	Months 4-5
5	Transfer Learning Fine-Tuning & Optimization	Months 5-6
6	Temporal Smoothing & API Integration	Month 6
7	Cross-Component Integration Testing	Months 6-7
8	Field Validation & Performance Evaluation	Months 7-8
9	Documentation & Final Submission	Months 8-9

8. RISK ANALYSIS

Risk	Impact	Mitigation
Insufficient Sri Lankan elephant audio data	High	Supplement with augmented data and transfer learning from African elephant datasets
Model misclassification in noisy conditions	High	Collect diverse background noise; apply noise augmentation during training
High audio processing latency	Medium	Optimize pipeline; use model quantization for faster inference
Microphone hardware failure in field	Medium	Use redundant microphone placement; implement hardware monitoring
Dataset class imbalance (aggressive events rare)	Medium	Apply SMOTE augmentation and class-weighted loss during training
API schema mismatch with Fusion component	Low	Define and freeze API schema early; use contract testing

9. REFERENCES

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